HIV-2 Protease-Inhibitor Resistance — Cross-Species Transfer Learning

Complete project flowchart: from origin (zero point zero) to the present submission.

Pin-to-pin record of every dataset, every curation step, every model decision, every evaluation and every revision. Numbers are those reported in the manuscript.

Data · Features · Modelling · Evaluation · Integrity findings · Manuscript & release · Review-driven revision

green = upheld · amber = qualified · red = negative / withdrawn

PHASE 0 — Problem definition — why this project exists

3 steps

0.1 Identify the clinical gap

HIV-2 infects roughly 1–2 million people. It is intrinsically resistant to several protease inhibitors, and only a subset (darunavir, lopinavir) is clinically active. Resistance interpretation therefore matters directly for second-line therapy.

0.2 Establish that existing tools are HIV-1-derived

Resistance-interpretation resources — the Stanford HIVDB penalty score and the rule-based HIV-2EU v4 algorithm — are expert-curated rule tables calibrated largely on HIV-1. They are *interpretation* systems, not learned predictors.

0.3 Verify the novelty gap by literature search

Systematic search (PubMed, arXiv, Scholar, bioRxiv/medRxiv) found **no machine-learning or deep-learning predictor of HIV-2 protease-inhibitor resistance**, and no HIV-1→HIV-2 cross-species transfer for resistance prediction.

- Adjacent work is HIV-1 only (Steiner 2020; Shen 2016; Pawar 2018; Hepler 2014; Farquhar 2026; Aggarwal & Periwal 2026) or rule-based for HIV-2 (HIV-2EU, Charpentier 2025).
- Closest same-corpus work: Farquhar 2026 uses the same Stanford PI corpus but stays within HIV-1.

Gap confirmed — this is the premise of the study.

PHASE 1 Data acquisition and curation

7 steps

1.1 Acquire the HIV-1 training corpus

Downloaded the Stanford HIVDB HIV-1 protease genotype–phenotype dataset via Entrez/HTTP (`PI_DataSet.tsv`), then expanded to per-drug assay rows.

- Raw download: **2,171 sequences**
- Expanded corpus (before cleaning): **7,183 rows**

1.2 Remove synthetic rows integrity finding

1,500 of the 7,183 rows were **not experimental measurements** but outputs of a banded-noise generator (`Stanford_Score ≥ 30 → U(−6.5,−4.5)`, else `U(−12.0,−9.5)`, seed 42). Because the generator used the very score the model would be asked to predict, these rows induce a self-fulfilling genotype–phenotype correlation.

- Removed and quarantined before any modelling.
- 7,183 − 1,500 = **5,683 genuinely measured HIV-1 assays**

1.3 Acquire the HIV-2 cross-species panel

Curated HIV-2 protease–phenotype assays from ChEMBL targets **CHEMBL380** and **CHEMBL5074**.

- Raw: 321 rows → 320 after drug-name mapping (one 'Experimental PI / Analogue' entry has no mapped drug)
- **137 rows** — zero-shot test set
- **160 rows** — fine-tune panel (4 drugs × 40)
- **19 unique sequences** — the true effective sample size

1.4 Assemble the HIV-2 clinical cohort

Two sources, both scored with Stanford HIVDB penalties so they are comparable:

- **Stanford_HIVDB** — 3,456 rows; sequences reconstructed from reported mutation lists.
- **Mined_Expanded** — 1,776 rows; HIV-2 protease sequences retrieved from **NCBI GenBank** by automated Entrez query `(db="protein", term="HIV-2 protease")`, trimmed and aligned to the 99-residue frame, then scored.

Total **654 sequence entries × 8 PIs = 5,232 rows**; the headline four-common-PI subset is **2,616 rows**.

1.5 Apply the common curation pipeline (seven steps)

- **(1) Length/frame filter** — keep exactly 99-residue sequences; removes fragments and mis-annotated ORFs that would correlate with source rather than resistance.
- **(2) Align to a shared frame** — HIV-2 to ROD, HIV-1 to HXB2, at 1:1 correspondence. Without this, position 82 in HIV-1 ≠ position 82 in HIV-2 and transfer is meaningless.
- **(3) Species by provenance, not sequence** — inferred labels would be circular for a study premised on species difference.
- **(4) Drug-name normalisation** — unify Stanford abbreviations and ChEMBL names; unmappable records dropped.
- **(5) Target construction** — IC50 (nM) → ΔG via `ΔG = RT·ln(IC50×10⁻⁹)`, RT = 0.593 kcal/mol at 298 K; retain $\Delta G \in [-20, 0]$.
- **(6) Redundancy removal + identity clustering** — 5,683 rows collapse to **1,882 unique sequences**; clustered at 95% identity → **882 clusters + 714 singletons**.
- **(7) Overlap audit** — 0 sequences shared between the clinical cohort and the HIV-1 pre-training corpus; 3 shared with the ChEMBL fine-tune panel.

> **Why step 6 matters.** Sequence clustering is not cosmetic data cleaning — it is what makes cross-validation honest. If near-identical sequences were split across folds, the model would effectively memorise an isolate and inflate every reported correlation. All within-species CV therefore uses **GroupKFold on these clusters**.

1.6 Quantify within-corpus diversity

- HIV-1 (HXB2) vs HIV-2 (ROD): **51 of 99 positions divergent → 48.5% identity**
- HIV-2 ChEMBL panel vs ROD wild-type: **88.9%–99.0% identity, mean 95.0%**; no sequence is 100% wild-type.

1.7 Lock inputs by hash

SHA-256 of every input file is recorded inside each result manifest, so any modification of the inputs is detectable and every reported number is traceable to a specific dataset state.

PHASE 2 **Feature engineering** 8 steps

2.1 Choose physicochemical amino-acid descriptors

Each residue is replaced by **21 numbers** describing its physical chemistry, drawn from three independently derived encodings:

- **Hellberg Z-scales (3)** — first 3 principal components of ~29 measured properties: Z1 hydrophobicity, Z2 steric bulk, Z3 electronic character.
- **Kidera factors (10, orthogonal)** — factor analysis of **188** physical properties. KF1 helix/bend, KF2 side-chain size, KF3 extended-structure, KF4 hydrophobicity, … KF10 surrounding hydrophobicity.
- **VHSE scales (8)** — principal components of 50+ properties; VHSE1 hydrophobic, VHSE2 steric, VHSE3 electronic.

Why these, and not structure or a language model? They are properties of the *amino acid itself*, and are therefore identical in HIV-1 and HIV-2 — which is precisely what makes cross-species transfer meaningful. They also need no experimental structure, and HIV-2 structures are scarce.

2.2 Build the feature matrix

- 21 descriptors × 99 positions = **2,079-D**
- + mean and SD of each descriptor across the 99 positions (+42) = **2,121-D** (locked deployment bundle)
- Scaled with a scaler fitted on HIV-1 only — no leaking of HIV-2 or clinical statistics.

2.3 Map descriptors to resistance mechanism

The three families correspond to the three physical routes by which a substitution weakens drug binding — this is the mechanistic justification for the feature choice:

Family	Physics	Mechanism	HIV-2 examples
Z2, VHSE2, KF2, KF6	Steric bulk / volume	Clash with the bound inhibitor	I50V, I54M, I84V
Z1, VHSE1, KF4, KF10	Hydrophobicity	Altered packing → weaker binding	pocket packing changes
Z3, VHSE3, KF9	Electronic / pK	Loss of H-bond or polar contact	polar-contact changes
KF1, KF3, KF5, KF7, KF8	Backbone preference	Flap flexibility gating the site	flap region 45–55

2.4 Define pocket position sets for masking experiments

Derived from HIV-2 ROD structure **3S45** (1.51 Å) by heavy-atom distance to the ligand, in the 1-based 99-residue frame:

- **Core pocket, 16 residues within 5.0 Å**: `8 23 25 27 28 29 30 31 32 47 48 49 50 81 82 84` — includes the catalytic dyad (25, 27) and resistance sites 50, 82, 84.
- **Pocket-19**: adds `26 45 46 54 76 80` → the 399-D masked set (21 × 19).
- **Extended 8.0 Å** shell; and **BASE-A** = 16 core + 18 conserved neighbours = 34 residues → 714-D.

Terminology fixed. Masking means *selecting positions*, not weighting residues. The pipeline does not assign greater weight to pocket positions; it simply restricts the feature matrix to them. Masking is used as a mechanistic probe, not as the deployment configuration.

2.5 Predict structures with ESMFold structural branch

Because no HIV-2 protease co-crystal exists with atazanavir, darunavir, lopinavir or saquinavir, structures were *predicted* rather than measured.

- **ESMFold** folded **23 HIV-2 unique sequences**; per-residue **pLDDT** confidence recorded for every model.
- The same procedure was applied at scale to the HIV-1 unique-sequence corpus.
- Experimental structures kept for validation only: **3S45** (1.51 Å, HIV-2) and the HIV-1 co-crystals **4LL3, 2AQU, 3OXC, 1MUI**. All superposition and pocket analysis uses these measured coordinates, never the predicted ones.

Naming note. These are **ESMFold** models, not AlphaFold. Separately, **ESM-2 650M embeddings** were used as a feature view and as a comparator (standalone R = 0.6759; blended 0.6994) — a different use of the same protein-language-model family.

2.6 Build protease–drug complexes by docking

To get a *structure–ligand* representation rather than a structure-only one, each protease was docked with each inhibitor using **AutoDock Vina** (exhaustiveness 16).

- HIV-2: **19 mutants × 9 inhibitors = 171 docked complexes**, of which **136** matched to measured ChEMBL rows
- HIV-1: **3,382 sequences × 4 inhibitors = 13,528 holo complexes**
- Three docking campaigns run (physics-based, refined, final Vina); Vina treated as authoritative

2.7 Extract ONION-Net features from every structure

Each apo and holo structure was passed to the **ONION-Net** protocol — 62 distance shells (0–31 Å) × 167 atom-type pairs describing each atom's local environment.

- apo form: **10,309-D**
- docked (ligand-embedded) form: **10,478-D**
- Extracted separately for HIV-1-derived and HIV-2-docked structures — the distinction that determines whether the features transfer

2.8 What the structural branch showed

negative result

All three structural results are negative for structure as a *deployed* representation: HIV-1-derived ONION-Net does not transfer (**R = −0.1392**); naive sequence+structure concatenation is *worse* than sequence alone (**0.2061 vs 0.3582**); and structure–ligand features help only when computed on *species-matched* HIV-2 docked complexes (corrected **R = 0.6694**), as a correction stage rather than a replacement.

Two cautions recorded. (1) **The evaluation partition matters more than the docking method**: the same 136 entries give R ≈ 0.79 under naive KFold but ≈ 0.67 under GroupKFold — the higher figure is a **leakage artefact** and is used nowhere in the paper. (2) ONION-Net features computed from *literature* structures map to **zero** ChEMBL entries, so every structural number rests on predicted, docked complexes, not experimental holo structures.

PHASE 3 **Modelling — training, model choice, and transfer protocol** 5 steps

3.1 Train within-HIV-1 models

Target is ΔG in kcal/mol (not raw IC50, which is assay-dependent). Random Forest (600 trees, depth 20) and XGBoost (800 trees, depth 6, lr 0.03) on the 2,121-D bundle, validated with **GroupKFold at 95% identity**.

Within-HIV-1 grouped CV: **R = 0.7387 ± 0.0472** — the internal reference against which transfer is judged.

3.2 Apply zero-shot to HIV-2

The HIV-1 model, with no HIV-2 fitting, is applied to HIV-2 ChEMBL rows.

- Global RF zero-shot: **R = 0.5163** (n = 137), transfer ratio **0.699** (= 0.5163 / 0.7387)
- Best zero-shot configuration (ridge meta-ensemble): R = 0.5818
- Strict same-feature-space baseline: R = 0.2492

Upheld — biophysical signal survives ~50% sequence divergence.

3.3 Fine-tune on HIV-2 — the actual transfer mechanism

No weights are modified, averaged or re-initialised. Transfer is performed by feature augmentation with a frozen model, in three steps:

- (i) the HIV-1 per-drug model is trained and then **frozen**;
- (ii) its ΔG prediction on each HIV-2 sequence becomes a single appended column — a **meta-feature**;
- (iii) a **new** Random Forest is trained per drug on `[2,121-D | meta-feature]`, with 5-fold out-of-fold pooling.

The deployment model is the new RF, not the HIV-1 model. Locked OOF benchmark: **R = 0.7018** (n = 160).

3.4 Attempt deep learning and record the failure negative result

A pocket graph neural network (PocketGNN) and a biophysical transformer (DBPT+) were trained. Both **failed**: R = −0.17 and 0.21 respectively.

Diagnosis: data starvation — 19 unique sequences cannot train a GNN. This is a statement about the available data, not a verdict against deep learning. Consistent with which, a *pre-trained* protein language model (ESM-2 650M) is competitive (standalone R = 0.6759; blended 0.6994) precisely because it arrives pre-trained.

Qualified — DL tiers are excluded from headline results and the exclusion is stated, not hidden.

3.5 Justify the final model choice

Tree ensembles are used because of **sample size**: greedy feature selection handles 2,121 partially redundant descriptors without dimensionality reduction; no scaling or architecture search is needed, so results are reproducible from a fixed seed; they are robust to uninformative inputs; and they expose feature importance, which enabled the pocket-importance check.

Independent validation of the feature choice: restricting to the 19 pocket positions, the top five residues by importance — `54, 84, 28, 48, 26` — carry **72.4%** of the importance mass. Positions 54 and 84 are the canonical resistance sites (I54M, I84V): the model recovered the mechanistic core from data alone.

Note masking does *not* improve over the global representation (0.5014–0.5036 vs 0.5163), so pocket restriction is a probe, not the deployment setting.

PHASE 4 **Evaluation — five independent lines of evidence** 6 steps

4.1 Head-to-head against rule-based interpretation primary evidence

The clinical evaluation target (HIVDB penalty) is itself a rule, so a separate, *non-circular* comparison was run against **measured** binding affinity: the Stanford-derived single- plus combination-mutation penalty tables applied to the same sequences.

Drug	n	model R	rule R	ΔR
ATV/r	40	0.6125	0.3907	+0.2218
DRV/r	40	0.8272	0.5406	+0.2866
LPV/r	40	0.5440	0.3929	+0.1511
SQV/r	40	0.7048	0.5624	+0.1424
Pooled	**160**	**0.7018**	**0.4724**	**+0.2294**

Sequence-clustered paired bootstrap, 2,000 resamples: ΔR 95% CI [0.060, 0.408], **P(ΔR ≤ 0) = 0.001**. The model is higher on every drug.

4.2 Test robustness to individual sequences

Because 160 rows contain only **19 unique sequences**, a leave-one-sequence-out jackknife was run.

Upheld — the model remains ahead in **100% of folds** (ΔR range [+0.160, +0.264]); the rule-blind signal stays positive in 100% of folds (R range 0.48–0.70). The result is not carried by any single sequence.

4.3 Quantify rule coverage strongest contribution

The rule engine reproduces the Stanford clinical penalty *exactly* (R = 1.0000; 1,728/1,728 rows), showing the clinical benchmark is rule-determined. Coverage was then measured directly:

- Clinical benchmark is **zero for 93.1%** of isolates (91.3–96.9% across 8 drugs)
- Stanford-derived rules recognise **3 of 37** mutations in the measured panel
- Rule is silent on **72.5%** of the measured panel
- On those 116 rule-blind rows the model retains **R = 0.585** (95% CI 0.296–0.788; P < 0.001) while the rule cannot rank them at all

4.4 Repeat coverage against the official tool

closes a key objection

To show the coverage finding is not an artefact of the re-implementation, the same panel was scored with the **official published HIV-2EU v4 rules** (Charpentier 2025):

- **83.1%** of the measured panel unscored (worse than the 72.5% baseline)
- **No protease rules at all for 6 of 8** clinical drugs (saquinavir withdrawn; atazanavir classified naturally resistant)
- Covers **5 of 37** observed mutations

Upheld — the official tool is *sparser* than the baseline used, so the reported comparison is the conservative one.

4.5 Clinical cohort evaluation

secondary — rule-derived target

The deployment model was applied, with no retraining, to **2,616 never-seen clinical rows** (654 entries × 4 PIs), after confirming zero sequence overlap with the HIV-1 corpus.

- Pooled Pearson R = **0.4454**; Spearman ρ = 0.3388; RMSE = 13.62
- ROC-AUC = **0.927**
- Correlation with known drug-resistance-mutation counts up to r = 0.79 (LPV/r)

Not claimed as superiority. Within every drug the same-space zero-shot baseline correlates more strongly, and the pooled advantage is a between-drug calibration effect rather than a within-drug accuracy gain.

4.6 Record a contradictory finding rather than hide it

unresolved

The model does not reproduce the literature-verified **I82F** darunavir hypersusceptibility override. Reported explicitly as a directional caveat (n = 5 I82F-only rows), not claimed as a result. Notably, I82F is one of the mutations the rule tables assign *no* penalty at all (see 4.3).

PHASE 5

Integrity findings — what the audit changed

4 findings

5.1 Synthetic training rows discovered and removed

1,500 generated rows found in the source corpus and quarantined before modelling (Phase 1.2). The quarantine file ships with the code so the exclusion is auditable.

5.2 Circularity of the clinical target quantified

The rule engine reproduces the HIVDB penalty exactly (R = 1.0000). The clinical benchmark therefore measures rule-consistency as much as biology — which is why the *measured* comparison (4.1) is treated as primary and the clinical evaluation (4.5) as supporting.

5.3 Reproducibility drift identified and bounded

The same per-drug values appeared differently in different tables (ATV/r 0.6123 vs 0.6125) because Table 8 came from a pipeline re-run that drifts in the 4th decimal from the locked manifest. Resolved by quoting the locked manifest everywhere and recording the reconciliation (max |drift| = **0.0002**) in the artefact itself.

5.4 Overlap audit

Confirmed 0 shared sequences between the clinical cohort and the HIV-1 pre-training corpus, and 3 with the ChEMBL fine-tune panel. No clinical label enters training at any stage.

PHASE 6 Manuscript and public release

3 steps

6.1 Manuscript assembled

16 tables, 9 figures, 22 references, structured as Introduction → Methods (six subsections) → Results (§3.1–3.8) → Discussion → Conclusion. Every reported number traces to a manifest.

6.2 Public repository released

`github.com/nunnajaswant-ship-it/HIV-2_DRUG_RESISTANCE_TRANSFER_LEARNING_ON_HIV-1` — 97 files: curated datasets, full pipeline, result manifests, figure renderers, manuscript source and compiled PDF, MIT licence, CITATION.cff, pinned requirements.

Integrity gates refuse to write output when a re-derived metric disagrees with the locked manifest beyond tolerance, and every figure reads its numbers live from the manifests so it cannot drift.

6.3 Unsourced values given provenance

Three ONION-Net table values traceable to no manifest were located in `POCKET_ONIONNET_PCA_REPORT.md` and `ABLATION_STUDY_REPORT.md`; both reports were added to the repository so all published numbers have a source.

PHASE 7 **Review-driven revision — new experiments and honest qualification** 8 steps

7.1 Seed sensitivity quantified

weakens headline

The locked benchmark used a single seed. Repeating the whole fine-tune + OOF under 10 random seeds gives **R = 0.628 ± 0.074** (range 0.516–0.715). The locked 0.7018 therefore sits at the *upper end* of the distribution, not its centre.

Qualified — the paper now reports the seed spread in the Results, abstract, conclusion and limitations, and states that comparisons are only made within a seed.

7.2 Fine-tuning-size sweep

negative result

Trained on k HIV-2 rows per drug (k = 1, 2, 5, 10, 20, 30; 20 splits each) with and without the HIV-1 meta-feature, testing whether the HIV-1 contribution grows as HIV-2 data shrinks.

k per drug	1	2	5	10	20	30
transfer R	0.017	0.102	0.330	0.492	0.606	0.641
no-HIV-1 R	0.017	0.100	0.315	0.490	0.607	0.641
gap	+0.000	+0.002	+0.015	+0.001	−0.002	−0.000

The HIV-1 contribution never exceeds 0.015 and does not grow as data is removed. HIV-2 data volume — not HIV-1 pre-training — drives performance. The transfer-learning framing is therefore reported as a demonstration that a compact physicochemical representation transfers in principle, not as a measurable gain at n = 160.

7.3 Pooled-statistic explanation tested, not asserted

Pooled Pearson R is governed by between-drug scale and is not a simple average of per-drug accuracy. Verified on the panel: raw pooled R = 0.701; after per-drug standardisation = 0.653. Every clinical claim is therefore stated per drug, with pooled values reported only alongside.

7.4 Unexplained data source documented

Mined_Expanded is now defined in Methods: HIV-2 protease sequences retrieved from NCBI GenBank by automated Entrez query, aligned to the 99-residue frame and scored with HIVDB penalties so they are directly comparable to the registry rows.

7.5 Cross-reference audit

A wrong table reference was found and fixed: the text cited "Tables 7, 9" for the zero-shot results, but Table 7 is training-data scaling and Table 9 is rule coverage; the correct target is Table 10. A full automated proofread now confirms every table, figure and section reference resolves.

7.6 Methods expanded to full detail

Methods grew from 15.8k to 30.5k characters to document: the seven-step curation with the validity threat each removes; the redundancy-removal and clustering rationale; per-descriptor mechanistic mapping (new Table 3); the model-choice justification and the DL failure; and the explicit transfer and masking protocol including all pocket positions.

7.7 Redundancy and length cleaned up

Two unfunded tool sections (conformal prediction, mutational scanner) were merged into one short subsection; the per-drug-superiority caveat was consolidated from twelve occurrences into one full statement plus pointers; unused material removed.

7.8 Figure 1 rebuilt

The original study-design figure was stale and incomplete. It was replaced by a four-stage design covering curation, features, model/transfer and evaluation — with all numbers read live from the manifests so it cannot drift again.

SUMMARY

Final state: what is claimed, what is qualified, what failed

read this last

Claim	Value	Status
First ML predictor of HIV-2 PI resistance	no prior work found	upheld
Within-HIV-1 grouped CV	R = 0.7387 ± 0.0472	upheld
Zero-shot HIV-1 → HIV-2 transfer	R = 0.5163, ratio 0.699	upheld
Beats rule-based interpretation on *measured* phenotypes	0.7018 vs 0.4724; ΔR = +0.229; P = 0.001	upheld
Robust to leave-one-sequence-out	ahead in 100% of folds	upheld
Rule-based interpretation is sparse	93.1% zero; HIV-2EU 83.1% unscored; 3 vs 5 of 37 mutations	upheld
Model works where the rule is silent	R = 0.585, P < 0.001	upheld
Headline single-seed benchmark	0.7018 (locked) vs 0.628 ± 0.074 across seeds	qualified
HIV-1 pre-training provides a measurable transfer gain	gap ≤ 0.015 at every data volume	not claimed
Per-drug clinical superiority	zero-shot baseline higher within every drug	not claimed
Deep-learning tiers	PocketGNN −0.17; DBPT+ 0.21	failed, reported

The through-line. The work began as an attempt to show that HIV-1 knowledge transfers to HIV-2. That claim is supported in principle (zero-shot transfer works) but *not* as a measurable gain at n = 160, and the honest audit reports this. What emerged as the strongest and most defensible contribution was not the model at all, but the quantification of how sparse the existing rule-based standard is: 93.1% of the clinical benchmark is zero, and the official HIV-2EU rules define no protease interpretation for six of eight drugs. The paper's value now rests on that finding, supported by the measured head-to-head, rather than on the transfer narrative.

Companion to the manuscript *HIV2_cross_species_transfer_manuscript.pdf* and the repository `github.com/nunnajaswant-ship-it/HIV-2_DRUG_RESISTANCE_TRANSFER_LEARNING_ON_HIV-1`. Every quantitative value above is reproduced from a result manifest in that repository.